

Protect the Decoder: Representation-Preserving Adaptation and Failure Detection for Far-OOD MedSAM

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Abstract. Parameter-efficient adaptation of MedSAM to one imaging domain can leave it worse than zero-shot on distant domains, a collapse that earlier work linked to drift of the mask decoder’s representations from the base model. We intervene on that mechanism: a centred kernel alignment (CKA) loss pulls the adapted decoder’s activations on a 32-image out-of-distribution probe back toward the base decoder during LoRA training on dermoscopy. Over three seeds the regulariser raises tight-box Dice by 0.05 on breast ultrasound and 0.10 on mammography with no in-domain cost; the same anchor on the image encoder alone lowers far-OOD Dice, and anchoring encoder and decoder together gives the largest gain. Varying the adaptation budget from 50 to 2595 images shows that LoRA falls below zero-shot far-OOD already at 250 images, whereas decoder-only and encoder-only adaptation never do. Finally, the per-image drift of the decoder’s upscaled mask embedding detects far-OOD failures of parameter-efficient adapters with AUROC 0.95 on ultrasound and 0.82 on mammography, where the decoder’s own IoU estimate is at chance, while the IoU estimate is informative for full fine-tuning. Far-OOD robustness of adapted MedSAM is decided at the decoder, and it can be both protected and monitored there.

Keywords: Medical image segmentation · Foundation models · Parameter-efficient fine-tuning · Out-of-distribution robustness · Failure detection

1 Introduction

MedSAM [12] adapts the Segment Anything Model [7] to medical images and is routinely fine-tuned again for a target modality. Parameter-efficient methods such as LoRA [5] and visual prompt tuning [6] make this cheap, but a prior comparison on a four-dataset drift ladder showed that the cheapest methods are also the most fragile: LoRA trained on dermoscopy gains in-domain accuracy and then loses 0.18 Dice relative to zero-shot MedSAM on mammography, while decoder-only and full fine-tuning keep their far-OOD accuracy [2]. The same study found that the loss correlates with how far the mask decoder’s representations move from the base model, measured by centred kernel alignment (CKA) [8], and that encoder-side movement does not predict it. That was a correlation over checkpoints; this paper asks whether it is a mechanism that can

be acted on. First, does regularising the decoder toward the base model during adaptation reduce far-OOD collapse, and is the decoder specifically the locus? We add a CKA loss on a 32-image out-of-distribution probe and compare hooks on the decoder, on the encoder, and on both. Second, does the collapse scale with the adaptation budget? We train three adapters on nested subsets of 50, 250 and 1000 images at a fixed step budget. Third, can an adapted model flag its own far-OOD failures at inference? We compare two label-free per-image signals: the IoU estimate SAM’s decoder already produces, and the drift of the decoder’s upscaled mask embedding relative to the base model on the same image and prompt.

Our contributions are:

1. a CKA-regularised LoRA with an OOD-only probe, evaluated over three seeds and five prompt-jitter levels, and a hook-placement ablation showing that encoder-only anchoring hurts, decoder anchoring helps, and anchoring both is best;
2. an adaptation-budget study showing that LoRA’s far-OOD loss appears at 250 training images and does not recover with more data, while decoder-only and encoder-only adaptation stay above zero-shot at every budget;
3. a per-image failure detector based on decoder drift that reaches AUROC 0.95 and 0.82 on the two far-OOD sets for LoRA, together with the finding that the decoder’s IoU head is uninformative for parameter-efficient adapters but informative for full fine-tuning.

2 Related Work

Adapting SAM to medical images. MedSAM fine-tunes SAM’s encoder and decoder on over a million image-mask pairs with box prompts [12]; SAM-Med2D adds encoder adapters and retraining the decoder interactively [3]. Downstream users adapt these models again to one modality with LoRA [5], prompt tokens [6], or by unfreezing the decoder. The prior study we build on compared seven such strategies on a dermoscopy to ultrasound to mammography ladder and found that LoRA and prompt tuning collapse at the far end while decoder-only and full fine-tuning do not [2]. We take its testbed and diagnosis as given and study interventions.

Feature distortion under fine-tuning. Fine-tuning a good pretrained extractor can distort its features and underperform linear probing out of distribution [9]; which layers to tune depends on the type of shift [11]; interpolating fine-tuned and zero-shot weights recovers robustness [15]. Those results concern classifiers read out by a linear head; a promptable segmenter has a second network between encoder and output, and our results place the loss and recovery of robustness there.

Representation similarity as a training signal. CKA [8] compares two sets of activations up to orthogonal transformation and isotropic scaling. We use its linear form as a differentiable loss between an adapted model and its frozen base on a fixed probe batch, an anchor to the model’s own initialisation rather than distillation from a teacher.

Failure detection for segmentation. Estimating segmentation quality without ground truth ranges from reverse classification accuracy [14] to recent benchmarks of confidence-based failure detectors for medical segmentation [16]. SAM’s decoder emits a scalar IoU estimate for each mask [7], but its behaviour after adaptation has not been examined. We evaluate it against a detector derived from the mechanism studied here.

3 Method

f_θ denotes MedSAM with image encoder E_θ and mask decoder D_θ , θ_0 its released weights and θ the weights during adaptation; prompts are single boxes.

3.1 CKA-regularised adaptation

For activation matrices $X \in \mathbb{R}^{n \times p}$ and $Y \in \mathbb{R}^{n \times q}$ with centred columns, linear CKA is

$$\text{CKA}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \|Y^\top Y\|_F} \in [0, 1], \quad (1)$$

invariant to orthogonal transformation and isotropic scaling of either side [8]. Let P be a fixed probe batch of n images with boxes and $A_\ell(\theta; P)$ the activation captured by a forward hook at layer ℓ , one row per probe image. With $A_\ell(\theta_0; P)$ cached once, the training objective is

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \sum_{\ell \in H} w_\ell (1 - \text{CKA}(A_\ell(\theta_0; P), A_\ell(\theta; P))), \quad (2)$$

where H is the set of hooked layers, w_ℓ per-layer weights and $\mathcal{L}_{\text{task}}$ the equally weighted sum of binary cross-entropy and soft Dice. The probe forward runs every fourth step with λ multiplied by four, which keeps the expected regularisation gradient per step and quarters its cost; CKA terms are computed in single precision. The probe is OOD-only: 16 breast-ultrasound and 16 mammography images drawn once from the training splits of those datasets and absent from every evaluation set, each with a synthetic centred box covering half the image. No in-domain images are included, because the task loss already governs in-domain behaviour and an in-domain anchor competes with it. We use $\lambda = 10$, chosen in preliminary single-seed runs over $\{0.1, 1, 10\}$.

3.2 Hook placement

Decoder hooks capture the output of the decoder’s upscaling block ($32 \times 256 \times 256$), the IoU prediction head and the mask logits, with $w_\ell = 0.8, 1.5$ and 1.0 , following the strength of the per-layer correlation reported in prior work [2]. *Encoder* hooks capture the neck output ($256 \times 64 \times 64$) and the outputs of the last two transformer blocks ($64 \times 64 \times 768$), each with $w_\ell = 1$. *Both* is the union. Every configuration backpropagates through the encoder, where the LoRA weights live; the positions differ only in which representation is anchored.

3.3 Adaptation-budget protocol

One fixed permutation of the training set (seed 0) supplies its first 50, 250 and 1000 images, so smaller budgets are prefixes of larger ones and every method sees identical images. Each run performs 6000 optimiser steps at batch size one under a cosine schedule, validating every 500 steps and keeping the best checkpoint; the full-data point is each method’s seed-0 checkpoint. Prompts are tight.

3.4 Per-image failure signals

For a test image x with box b , two label-free signals are available. The first is the decoder’s IoU estimate $\hat{u}(x, b)$, a scalar SAM predicts alongside each mask [7] and that adaptation updates with the rest of the decoder. The second is the decoder drift

$$d(x, b) = 1 - \text{CKA}(U_{\theta_0}(x, b), U_\theta(x, b)), \quad (3)$$

where $U(x, b) \in \mathbb{R}^{65536 \times 32}$ is the decoder’s upscaled mask embedding for that image and prompt, with the 256×256 positions as samples and the 32 channels as features. It costs one extra base decoder forward per image, plus a base encoder forward when the encoder was adapted. A failure is a prediction with Dice below 0.5; both signals are scored by AUROC and AUPRC (high drift or low $\hat{u}$ positive), and $\hat{u}$ is compared with the true IoU by a ten-bin expected calibration error.

4 Experimental Setup

Drift ladder. All adaptation uses the ISIC 2018 dermoscopy training set [4] (2594 training, 100 validation, 1000 test images). Evaluation adds PH2 [13] (200 dermoscopy images from another clinic, near-OOD), BUSI [1] (647 benign and malignant breast-ultrasound images, far-OOD) and the 362 CBIS-DDSM test mammograms [10] with mass and calcification masks merged per image (far-OOD). Images are resized to 1024×1024 with SAM’s normalisation; box prompts come from the ground-truth mask, tight at evaluation unless stated.

Table 1. LoRA with and without the decoder CKA loss, both trained with 20 px jitter, tight-box evaluation, mean $\pm$ std over three seeds; Δ HD95 in pixels. p : paired Wilcoxon over per-image Dice at 20 px evaluation jitter, pairs matched on seed and image, Holm-corrected.

Dataset	no-CKA Dice	CKA Dice	Δ Dice	Δ HD95 (px)	p (Holm)
ISIC (ID)	0.949 ± 0.002	0.952 ± 0.002	$+0.003$	-0.3	<0.001
PH2 (near-OOD)	0.951 ± 0.001	0.954 ± 0.002	$+0.003$	-0.4	<0.001
BUSI (far-OOD)	0.774 ± 0.032	0.826 ± 0.008	$+0.051$	-96.1	<0.001
CBIS-DDSM (far-OOD)	0.437 ± 0.095	0.541 ± 0.095	$+0.104$	-152.4	<0.001

Methods. The seven strategies are those of the prior study [2]: zero-shot MedSAM; decoder-only fine-tuning; VPT-shallow and VPT-deep (ten learnable vectors at the encoder input or at every block, plus a trainable decoder); LoRA with rank 8 on the fused query-key-value projection of all twelve encoder blocks plus a trainable decoder; encoder-only LoRA with rank 28 on every encoder linear layer and a frozen decoder; and full fine-tuning. All use AdamW with a cosine schedule and mixed precision, with learning rates 5×10^{-4} (LoRA, prompts), 10^{-4} (decoder-only) and 10^{-5} (full). Regularised runs are LoRA plus Eq. (2). Training regime pm20 jitters each box corner uniformly within ± 20 pixels, pm0 uses tight boxes, and rand100 draws the jitter magnitude uniformly from $[0, 100]$ per step. A run takes six epochs on one 48 GB GPU.

Evaluation. We report Dice, IoU and HD95 (pixels on the 1024 grid). Prompt robustness expands each side of the tight box outward by an amount drawn uniformly from $[0, p]$ pixels, $p \in \{20, 50, 100, 200\}$, with one draw per image seeded on its identifier so all models see identical boxes. Multi-seed comparisons use seeds 0 to 2 and a paired Wilcoxon signed-rank test over per-image Dice, pairs matched on seed and image, Holm-corrected over the four datasets.

Prior results. At tight boxes the prior study reported zero-shot Dice of 0.823 and 0.692 on BUSI and CBIS-DDSM, LoRA at 0.780 and 0.509 (below zero-shot) against decoder-only and full fine-tuning at 0.894 to 0.901 and 0.828, and decoder-side CKA to the base correlating with far-OOD Dice at $\rho = 0.76$ against $\rho = 0.09$ for the encoder [2]. These baselines are not re-derived here.

5 Results: CKA Regularisation

Multi-seed effect. Table 1 gives the main comparison. The decoder CKA loss leaves in-domain and near-OOD Dice within 0.003 and raises far-OOD Dice by 0.051 on BUSI and 0.104 on CBIS-DDSM, with HD95 falling by 96 and 152 pixels: the regulariser removes the boundary failures of unregularised LoRA rather than small overlap losses. The BUSI gain is stable across seeds (std 0.008 against 0.032 without the loss); on CBIS-DDSM the seed spread is large for

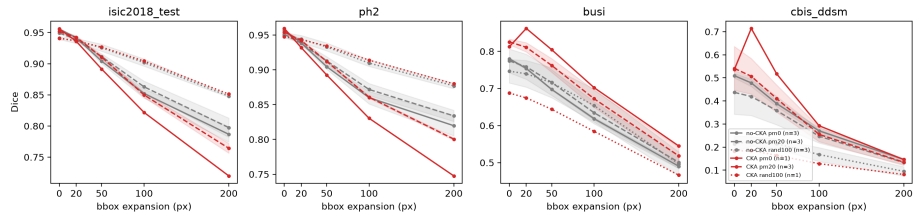


Fig. 1. Dice against outward box expansion for LoRA with (red) and without (grey) the decoder CKA loss under three training regimes: tight (solid), 20 px jitter (dashed), random jitter up to 100 px (dotted). Bands: one std over three seeds where available.

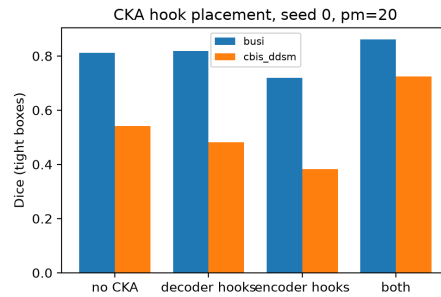


Fig. 2. Hook placement (LoRA, 20 px training, tight-box evaluation, seed 0): far-OOD Dice with no regulariser, decoder hooks, encoder hooks, and both. [Three-seed error bars follow once the remaining runs finish.](#)

both models (std 0.095) but the gain holds in every seed, and the paired test is significant on all four datasets with a median per-image gain of 0.088 on CBIS-DDSM. With tight-box training the loss gives smaller same-signed gains (0.033, 0.027; seed 0); with rand100 training it hurts (0.059 and 0.032 lower), so the two regularisers compound at mild jitter and conflict under heavy jitter.

Prompt robustness. In Fig. 1 the regularised pm20 model is the best of the six on both far-OOD sets at every expansion, with an advantage that narrows as boxes loosen: the paired gain on BUSI is 0.053, 0.046, 0.038 and 0.017 at 20, 50, 100 and 200 px, and on CBIS-DDSM 0.088, 0.052, 0.008 and not significant. On the dermoscopy sets the sign flips beyond 50 px (0.013 lower at 100 px, 0.033 at 200 px, $p < 10^{-20}$): anchoring to a base trained with tight boxes inherits its sensitivity to loose boxes in-domain. The rand100 regime is the other side of the trade, buying in-domain prompt robustness at a far-OOD price that the CKA loss makes worse.

Where the anchor must be. Figure 2 separates the two halves of the model. With the loss on encoder activations alone, far-OOD Dice falls below the unregularised baseline, from 0.811 to 0.720 on BUSI and 0.542 to 0.383 on CBIS-DDSM, at

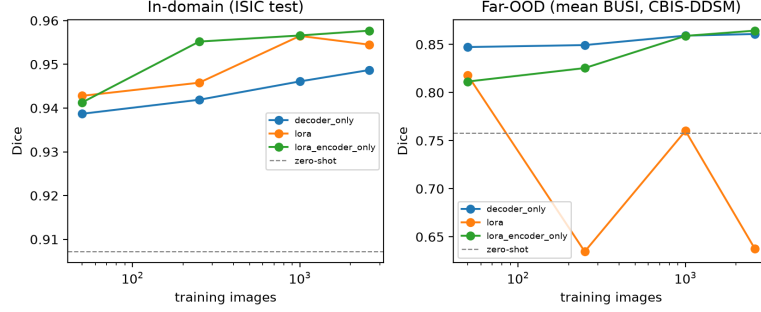


Fig. 3. Dice against training images (log axis) for three adapters trained for a fixed 6000 steps; dashed: zero-shot. Left: in-domain (ISIC test). Right: far-OOD, mean of BUSI and CBIS-DDSM. Single seed.

unchanged in-domain Dice: holding the encoder’s late features near the base does not protect the far-OOD output. Decoder hooks give the gains of Table 1. Anchoring both is best by a wide margin at seed 0, 0.861 on BUSI and 0.725 on CBIS-DDSM, the latter above zero-shot for the first time for any LoRA variant in this testbed, at no in-domain cost (0.950 on ISIC). *Seeds 1 and 2 of the encoder and both arms are in training; their mean and std replace this sentence.* The decoder is the locus, and the encoder anchor helps only once the decoder is anchored too, presumably because a stable decoder can exploit stable inputs instead of compensating for shifting ones.

6 Results: Adaptation Budget

Figure 3 shows how the collapse depends on how much of the target domain the adapter sees. Decoder-only fine-tuning and encoder-only LoRA improve on both axes with every increase in budget and stay above the zero-shot far-OOD reference of 0.758 throughout (0.847 to 0.861 and 0.811 to 0.864). LoRA with a trainable decoder is the only method that crosses it: level with the others at 50 images (0.818), it falls to 0.635 at 250, partly recovers to 0.760 at 1000 and ends at 0.638 with all 2595 images, while its in-domain Dice rises monotonically to the best of the three. The collapse is thus not a symptom of too little data or of repeated passes over the same images, since every budget receives 6000 steps; it appears as soon as a few hundred in-domain images reach a model whose decoder is free to move, and more data does not undo it. The non-monotonic shape between 250 and 1000 images is a single-seed observation that we do not interpret.

Table 2. AUROC for detecting per-image failures (Dice below 0.5) on the far-OOD sets at 50 px box jitter: decoder drift (Eq. (3)) versus the decoder’s IoU estimate. Cells with fewer than 20 failures are omitted; drift is zero by construction for zero-shot.

Method	BUSI (far-OOD)		CBIS-DDSM (far-OOD)	
	drift	IoU head	drift	IoU head
Zero-shot	—	0.87	—	0.71
Decoder-only FT	0.79	0.48	0.55	0.58
VPT-shallow	0.92	0.49	0.71	0.36
VPT-deep	0.93	0.63	0.61	0.59
LoRA	0.91	0.69	0.76	0.44
Encoder-only LoRA	0.35	0.89	0.36	0.75
Full FT	0.23	0.87	0.32	0.79

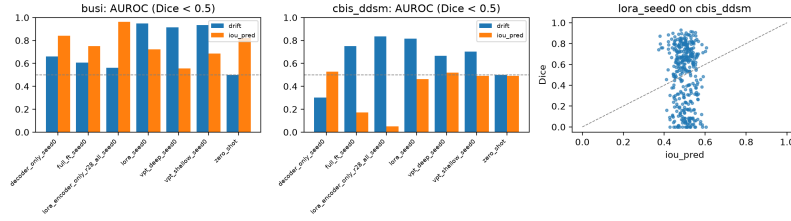


Fig. 4. Tight boxes. Left, centre: AUROC per method for drift (blue) and the IoU estimate (orange) on BUSI and CBIS-DDSM; dashed: chance. Right: IoU estimate against true Dice for LoRA on CBIS-DDSM; the estimate stays near 0.5 whether the mask is right or wrong.

7 Results: Failure Self-Detection

Drift detects the failures of parameter-efficient adapters. At tight boxes (Fig. 4) LoRA’s 61 failures among 647 BUSI images and 152 among 362 CBIS-DDSM images are detected with AUROC 0.95 and 0.82 (AUPRC 0.75 and 0.83 against failure rates of 0.09 and 0.42), VPT-shallow’s with 0.93 and 0.70, VPT-deep’s with 0.92 and 0.67. Under 50px jitter (Table 2) the ordering is unchanged, with drift at 0.91 to 0.93 on BUSI and 0.61 to 0.76 on CBIS-DDSM. Their IoU estimate is at or below chance (0.44 to 0.69), and the scatter in Fig. 4 shows why: after adaptation it collapses to a narrow band around 0.5 regardless of mask quality. Its calibration error on CBIS-DDSM is 0.13 for LoRA at tight boxes and 0.31 at 50px, where it predicts an IoU of 0.50 on average against a true IoU of 0.20.

The IoU estimate detects the failures of full fine-tuning. The pattern inverts for the methods that keep their far-OOD accuracy. Full fine-tuning and encoder-only LoRA fail on only 2 to 14 images at tight boxes, too few to score, but under 50px jitter their 220 to 236 CBIS-DDSM failures are detected by the IoU estimate with

AUROC 0.75 to 0.79 (0.87 to 0.89 on BUSI) while drift is below chance (0.23 to 0.36); zero-shot MedSAM is likewise served by its IoU head (0.87, 0.71). The two signals therefore cover different mechanisms: drift flags predictions on which the adapted decoder has left the representation region the base model would use, which is how parameter-efficient adapters fail, and it is uninformative once the decoder is retrained wholesale and moves for every image, exactly the case in which the IoU head stays usable. For a deployed adapter the rule is simple: few adapted decoder parameters, monitor drift; a fully retrained decoder, monitor its own IoU estimate.

8 Limitations and Conclusion

The ladder is asymmetric: all models are adapted on dermoscopy and tested toward ultrasound and mammography, and we have not shown that the anchors help when a far-OOD set is the training domain. The budget study is single-seed. The CKA loss adds about 30 percent to training time and the drift detector doubles inference cost when the encoder is adapted. The hook-placement result rests on [three seeds for the decoder arm and one for the other arms until the remaining runs complete](#). The regularised model inherits the base’s sensitivity to loose boxes in-domain, so it suits tight or lightly jittered prompts and deployment far from the training domain rather than every setting.

Within those bounds the evidence points one way. Far-OOD failure of adapted MedSAM arises where the mask decoder’s representations leave the base model’s; an anchor there removes much of the failure at no in-domain cost, an anchor on the encoder alone makes it worse, and the size of that departure on a single image predicts whether the prediction has failed. Protecting the decoder during adaptation and monitoring it at inference are two uses of one quantity, both cheap enough to be default practice when a medical foundation model is adapted in one domain and used in another.

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